LAB EXPERIMENT

COURSE CODE: ITA0318

COURSE NAME: MOBILE COMPUTING

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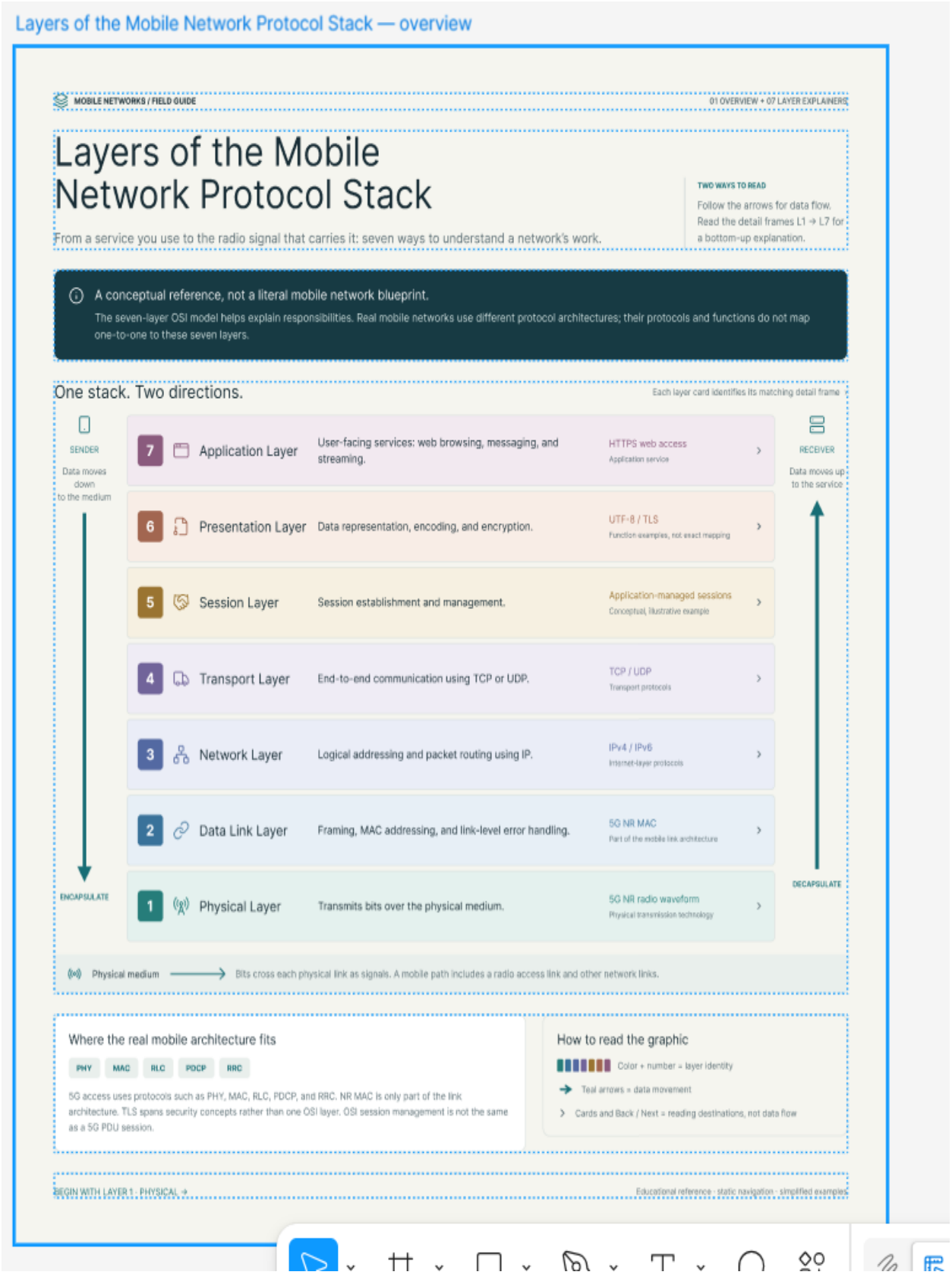
EXP -04

4. Create a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

Aim: -
Design a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

Procedure: -
1. openFigma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
11. Review and edit the Prototype
12. Save and Share

Design: -


Layers of the Mobile Network Protocol Stack — overview
MOBILE NETWORKS / FIELD GUIDE
01 OVERVIEW + 07 LAYER EXPLAINERS
Layers of the Mobile Network Protocol Stack
From a service you use to the radio signal that carries it: seven ways to understand a network's work.
TWO WAYS TO READ
Follow the arrows for data flow. Read the detail frames L1 → L7 for a bottom-up explanation.
A conceptual reference, not a literal mobile network blueprint.
The seven-layer OSI model helps explain responsibilities. Real mobile networks use different protocol architectures; their protocols and functions do not map one-to-one to these seven layers.
One stack. Two directions.
Each layer card identifies its matching detail frame
SENDER
Data moves down to the medium
ENCAPSULATE
RECEIVER
Data moves up to the service
DECAPSULATE
7 Application Layer
User-facing services: web browsing, messaging, and streaming.
HTTPS web access
Application service
6 Presentation Layer
Data representation, encoding, and encryption.
UTF-8 / TLS
Function examples, not exact mapping
5 Session Layer
Session establishment and management.
Application-managed sessions
Conceptual, illustrative example
4 Transport Layer
End-to-end communication using TCP or UDP.
TCP / UDP
Transport protocols
3 Network Layer
Logical addressing and packet routing using IP.
IPv4 / IPv6
Internet-layer protocols
2 Data Link Layer
Framing, MAC addressing, and link-level error handling.
5G NR MAC
Part of the mobile link architecture
1 Physical Layer
Transmits bits over the physical medium.
5G NR radio waveform
Physical transmission technology
Physical medium
Bits cross each physical link as signals. A mobile path includes a radio access link and other network links.
Where the real mobile architecture fits
PHY
MAC
RLC
PDCP
RRC
5G access uses protocols such as PHY, MAC, RLC, PDCP, and RRC. NR MAC is only part of the link architecture. TLS spans security concepts rather than one OSI layer. OSI session management is not the same as a 5G PDU session.
How to read the graphic
Color + number = layer identity
Teal arrows = data movement
Cards and Back / Next = reading destinations, not data flow
BEGIN WITH LAYER 1 : PHYSICAL →
Educational reference · static navigation · simplified examples

Physical Layer — explanation

MOBILE NETWORKS / FIELD GUIDE

Overview · all seven layers

L1 Physical | L2 Data Link | L3 Network | L4 Transport | L5 Session | L6 Presentation | L7 Application

LAYER 1

Physical Layer

Turn bits into signals that can travel.

01

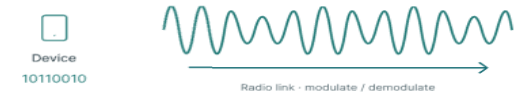
WHAT THIS LAYER DOES

The Physical Layer transmits bits over a physical medium. On a mobile radio link, it turns a stream of bits into a waveform and recovers bits from the received signal.

- **Represent bits as signals**
  Use modulation to carry information on a radio waveform.
- **Use the physical medium**
  Transmit and receive electromagnetic energy through antennas.
- **Recover the bit stream**
  Use timing, detection, and physical-layer coding to recover data.

Bits → waveform → bits

ILLUSTRATIVE FLOW



UNIT OF INFORMATION

Bits carried by physical signals

Radio signals travel across the air interface. The receiver processes the waveform to recover the transmitted information; the drawing is schematic, not a measured signal.

EXAMPLE **5G NR radio waveform**
A physical transmission technology, not an entire seven-layer stack.

IN A MOBILE NETWORK

5G New Radio (NR) PHY handles radio transmission and reception between a device and a base station. MAC schedules radio resources above it; the PHY carries out the physical transmission.

Remember: a bit is information; a waveform is how that information crosses a radio link.

BACK Overview

Layer 1 of 7 · read from L1 to L7

NEXT L2 · Data Link →

OSI is a conceptual reference; real mobile stacks do not map one-to-one.

Reading controls are static · arrows in diagrams show data flow

Data Link Layer — explanation

MOBILE NETWORKS / FIELD GUIDE

Overview · all seven layers

L1 Physical | L2 Data Link | L3 Network | L4 Transport | L5 Session | L6 Presentation | L7 Application

LAYER 2

Data Link Layer

Organize data for one link at a time.

02

WHAT THIS LAYER DOES

The Data Link Layer groups bits into frames and manages delivery across a local link. Its conceptual responsibilities include MAC addressing, access to the medium, and link-level error handling.

- **Create frame boundaries**
  Package higher-layer data into units a link can carry.
- **Identify link participants**
  Use link-specific addressing and medium-access rules.
- **Handle local errors**
  Detect errors and support recovery where the link provides it.

Package for a local link

ILLUSTRATIVE FLOW

Frame
Add link information
→
Send on link
Coordinate access
→
Check
Inspect received data

CONCEPTUAL FRAME · NOT AN NR FIELD LAYOUT



*If provided by the link technology.

A conceptual frame adds link information around a payload. Actual fields, checks, and recovery mechanisms depend on the link technology.

EXAMPLE **5G NR MAC**
Part of the mobile link architecture, not a complete or strict OSI Layer 2 mapping.

IN A MOBILE NETWORK

In 5G NR, MAC multiplexes data and manages access to radio resources. RLC and PDCP also perform link-related functions. NR uses radio-specific identifiers and procedures, not Ethernet-style MAC addressing on the air interface.

Remember: local-link delivery is different from routing a packet across many networks.

BACK L1 · Physical

Layer 2 of 7 · read from L1 to L7

NEXT L3 · Network →

OSI is a conceptual reference; real mobile stacks do not map one-to-one.

Reading controls are static · arrows in diagrams show data flow

Transport Layer — explanation

MOBILE NETWORKS / FIELD GUIDE

Overview · all seven layers

L1 Physical | L2 Data Link | L3 Network | L4 Transport | L5 Session | L6 Presentation | L7 Application

LAYER 4

Transport Layer

Carry data between application endpoints.

04

WHAT THIS LAYER DOES

The Transport Layer supports end-to-end communication between processes. TCP and UDP both use ports to identify application endpoints, but offer different delivery behavior.

- **Deliver to the right process**
  Port numbers distinguish application endpoints on a host.
- **Choose delivery semantics**
  TCP provides a reliable, ordered byte stream; UDP carries datagrams.
- **Manage end-to-end behavior**
  TCP uses acknowledgments, retransmission, flow control, and congestion control.

End-to-end, not just one hop

ILLUSTRATIVE FLOW

Sending app
Source port

→

IP network
Carries the packets

→

Receiving app
Destination port

TCP
Reliable, ordered byte stream

UDP
Datagrams; no delivery guarantee

The transport endpoints are the communicating processes. Intermediate IP routers normally forward packets without terminating the end-to-end transport connection.

EXAMPLE **TCP / UDP**
UDP itself does not guarantee delivery or order. Applications can add their own mechanisms.

IN A MOBILE NETWORK

Mobile applications may use TCP, UDP, or protocols built on them. For example, QUIC runs over UDP and adds reliable streams and security. Radio-link recovery and end-to-end transport recovery solve different problems.

Remember: reliable delivery is a protocol choice, not a promise made by every transport.

BACK
← L3 · Network

Layer 4 of 7 · read from L1 to L7

NEXT
L5 · Session →

OSI is a conceptual reference; real mobile stacks do not map one-to-one.

Reading controls are static · arrows in diagrams show data flow

Network Layer — explanation

MOBILE NETWORKS / FIELD GUIDE

Overview · all seven layers

L1 Physical | L2 Data Link | L3 Network | L4 Transport | L5 Session | L6 Presentation | L7 Application

LAYER 3

Network Layer

Address packets. Route them across networks.

03

WHAT THIS LAYER DOES

The Network Layer gives packets logical source and destination addresses. Routers examine the destination and forward each packet toward the next network on its path.

- **Use logical addresses**
  IP addresses identify endpoints independently of a single local link.
- **Choose a next hop**
  Routing information guides forwarding toward the destination.
- **Connect different networks**
  Carry packets over a sequence of underlying links.

One destination, multiple hops

ILLUSTRATIVE FLOW

Device
Create an IP packet

→

IP router
Forward to a next hop

→

Server
Receive the packet

ILLUSTRATIVE IPv6 DESTINATION · DOCUMENTATION ADDRESS
2001:db8::20 → forward toward the destination

The IP destination guides forwarding at each hop. Link details change along the path. This simplified drawing omits tunnels and intermediate nodes.

EXAMPLE **IPv4 / IPv6**
IP provides addressing and best-effort packet delivery; it does not guarantee arrival or order.

IN A MOBILE NETWORK

Mobile user traffic commonly carries IP packets through the radio access network and mobile core toward a data network. Mobility procedures and tunnels support delivery while the device's access connection changes.

Remember: IP answers "where should this packet go?", not "did the application receive it?"

BACK
← L2 · Data Link

Layer 3 of 7 · read from L1 to L7

NEXT
L4 · Transport →

OSI is a conceptual reference; real mobile stacks do not map one-to-one.

Reading controls are static · arrows in diagrams show data flow

LAYER 6

Presentation Layer

Make data understandable—and protect its form.

06

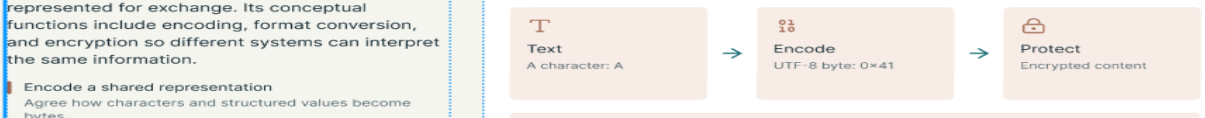
WHAT THIS LAYER DOES

The Presentation Layer describes how data is represented for exchange. Its conceptual functions include encoding, format conversion, and encryption so different systems can interpret the same information.

- **Encode a shared representation**
  Agree how characters and structured values become bytes.
- **Transform the data format**
  Convert or serialize information into an agreed form.
- **Apply security transformations**
  Encryption protects content; the receiver decrypts it for use.

Same meaning, different form

ILLUSTRATIVE FLOW



AT THE RECEIVER · REVERSE THE TRANSFORMATIONS

Decrypt → decode UTF-8 → recover the character A

This schematic shows encoding followed by protection. At the receiver, data is decrypted and decoded. Encryption does not replace agreement on a data format.

EXAMPLE **UTF-8 / TLS**

UTF-8 illustrates representation; TLS illustrates security functions. Neither is a strict OSI layer-equivalent mapping.

IN A MOBILE NETWORK

Mobile apps use encodings such as UTF-8 and security protocols such as TLS. TLS spans security concepts and is not confined to a literal OSI Presentation Layer. Radio-access security and application security also operate at different scopes.

Remember: representation makes bytes interpretable; encryption makes their contents confidential.

LAYER 5

Session Layer

Keep a conversation organized over time.

05

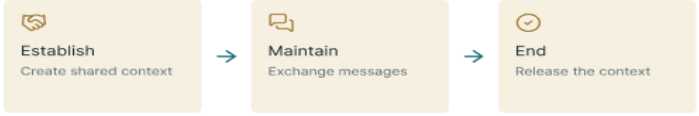
WHAT THIS LAYER DOES

The Session Layer is the OSI concept for establishing, managing, and ending a dialogue between applications. In common Internet stacks, these responsibilities often live inside application software.

- **Establish a dialogue**
  Create shared context so participants can begin an exchange.
- **Maintain session state**
  Track the conversation and coordinate its continuing activity.
- **End or resume an exchange**
  Close the dialogue or use application-defined recovery and resumption.

A dialogue has a lifecycle

ILLUSTRATIVE FLOW



ILLUSTRATIVE APPLICATION STATE

Conversation opened → context maintained → conversation closed

An application may keep session context across many messages. The steps represent conversation state, not the path taken by a single network packet.

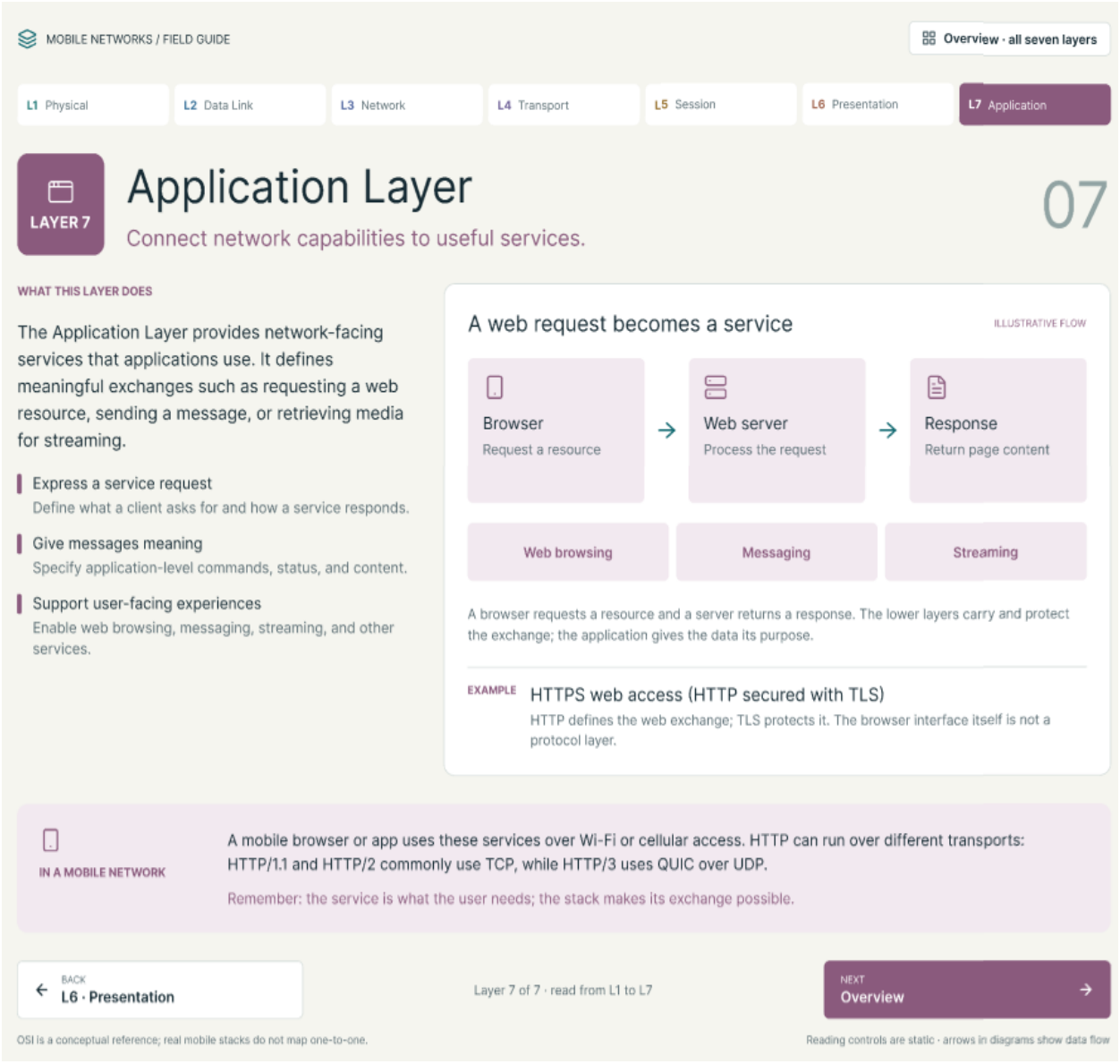
EXAMPLE **Application-managed sessions**

An illustrative function, not a claim that mobile networks use a dedicated OSI Session protocol.

IN A MOBILE NETWORK

A mobile messaging or collaboration app may manage its own conversation state. A 5G PDU session is a mobile-core connectivity construct; despite the shared word "session," it is not the OSI Session Layer.

Remember: an application session and a 5G PDU session are different concepts.



Result: -

Hence the visual representation in Figma highlighting the key elements of mobile network protocol stacks is created and executed successfully